DEPARTMENT OF TECHNICAL EDUCATION

GOVERNMENT OF KARNATAKA



450 - R.E.S. POLYTECHNIC, BANGALORE

SYSTEM AND NETWORK ADMINISTRATION ACTIVITIES

Course Code: 25CS34I | Semester III | Computer Science and Engineering

Submitted By:

Name: _____

Register Number: _____

RUBRICS FOR ASSESSMENT OF ACTIVITIES AND RECORD BOOK

Sl. No.	Dimension	Unsatisfactory (1-10)	Need Improvement (11-20)	Satisfactory (21-30)	Good (31-40)	Excellent (41-50)	Student Score
01	Understanding Activity Question	Do not understand at all	Poor understanding	Little understanding	Good understanding	Excellent understanding	
02	Execution of Activities	No	Major errors	Inaccuracy	Effective but minor errors	Precise and accurate	
03	Record Book Maintenance	No quality	Major flaws	Acceptable	Good quality	High quality	
04	In time submission of record	Major sections not addressed / poor	Takes time, 20% submitted in time	50% submitted in time	80% submitted in time	100% submitted in time	
Average Marks							

SYSTEM AND NETWORK ADMINISTRATION

One Activity per Week (Week 1 to Week 13) - Aim, Procedure and Result

Name: _____ Reg. No.: _____ Section: _____

WEEK 1: Computer Hardware Categories and Their Usage

Activity 1: Identify the front and rear panel controls and ports on a PC cabinet

Aim:

To identify the front and rear panel controls and ports of a PC cabinet.

Procedure:

1. Place the PC cabinet on a table with the power cable disconnected.
2. Observe the front panel and note the power button, reset button, USB ports, audio jacks and LED indicators.
3. Observe the rear panel and note the USB, HDMI/VGA/DisplayPort, LAN (RJ45), audio ports and the power supply socket.
4. Write the name and use of each control and port in a table and draw a labelled sketch.

Result:

The front and rear panel controls and ports were identified and their uses noted.

WEEK 2: Power Supply Unit - SMPS

Activity 2: Measure SMPS voltage levels using a multimeter

Aim:

To identify the SMPS output voltages by colour code and measure them using a multimeter.

Procedure:

1. Remove the SMPS from the cabinet and note the colour of the wires on the 24-pin connector.
2. Switch on the SMPS by shorting the green wire (PS_ON) with a black wire (ground) using a paper clip. Keep the SMPS switched on only for the test.
3. Set the multimeter to DC voltage and place the black probe on a black wire.
4. Place the red probe on the yellow, red, orange and blue wires one by one and note the readings.
5. Compare the readings with the standard values (within +/- 5 %) and switch off the SMPS.

Result:

Voltages were measured. Standard values: yellow +12 V, red +5 V, orange +3.3 V, blue -12 V, black ground, green PS_ON.

WEEK 3: Exploring Motherboard

Activity 3: Find the CPU features and cache details using the CPUID CPU-Z tool

Aim:

To find the features of the CPU and its cache using the CPU-Z tool.

Procedure:

1. Download and install CPUID CPU-Z from the official website.
2. Open CPU-Z and open the CPU tab; note the processor name, number of cores, threads, clock speed and socket.
3. Open the Caches tab and note the L1, L2 and L3 cache sizes.
4. Open the Mainboard and Memory tabs and note the motherboard model and RAM details.
5. Record all the values in a table.

Result:

The CPU features and cache details of the working system were identified using CPU-Z.

WEEK 4: Chipset Architecture and Booting Process

Activity 4: Check system information in BIOS/UEFI and change the boot order

Aim:

To check the system information in BIOS/UEFI and change the boot order.

Procedure:

1. Restart the computer and press the BIOS key (F2, F10 or Del, depending on the make) during start-up.
2. In the Main or System Information tab, note the BIOS version, processor, memory and serial number.
3. Open the Boot tab and move the required device (for example USB) to the first position using the arrow or +/- keys.
4. Save the changes and exit (F10) and observe that the system boots from the selected device.

Result:

The system information was checked and the boot order was changed successfully.

WEEK 5: Operating System Installation

Activity 5: Install and boot Windows or Linux

Aim:

To create a bootable device and install an operating system (Windows 10/11 or Linux).

Procedure:

1. Download the OS ISO file and create a bootable pen drive using Rufus or the Media Creation Tool.
2. Insert the pen drive, open the boot menu and boot from it.
3. Select the language, click Install and choose the partition for installation.
4. Wait for the files to be copied and the system to restart automatically.
5. Create the user account, set the region and network and complete the setup.
6. Log in, check the desktop and document the steps.

Result:

The operating system was installed and booted successfully.

WEEK 6: Windows Command Line Interface and User Management

Activity 6: Create and manage user accounts using PowerShell

Aim:

To create, modify and delete a user account on Windows using PowerShell.

Procedure:

1. Open PowerShell as Administrator.
2. Create the user: New-LocalUser -Name TestUser -Password (ConvertTo-SecureString 'Pass@123' -AsPlainText -Force).
3. Add to the Administrators group: Add-LocalGroupMember -Group Administrators -Member TestUser.
4. Verify using Get-LocalUser and log in with the new account.
5. Delete the account using Remove-LocalUser -Name TestUser and document the steps.

Result:

The user account was created, verified by login, and deleted using PowerShell.

WEEK 7: OS Configuration and Administration

Activity 7: Solve system start-up issues using Safe Mode

Aim:

To solve system start-up issues using Safe Mode.

Procedure:

1. Open Settings, go to Recovery and click Restart now under Advanced start-up.
2. Select Troubleshoot, Advanced options, Startup Settings and Restart.
3. Press 4 or F4 to start in Safe Mode.
4. In Safe Mode uninstall the faulty program or driver, or run System Restore.
5. Restart normally and check that the problem is solved.

Result:

The system was started in Safe Mode and the start-up problem was solved.

WEEK 8: Basics of File Systems and Disk Management

Activity 8: Create and format a new partition

Aim:

To create and format a new partition using Disk Management.

Procedure:

1. Right-click the Start button and open Disk Management.
2. Right-click the unallocated space (or shrink an existing volume to create it) and select New Simple Volume.
3. Enter the volume size, assign a drive letter and choose the file system (NTFS).
4. Enter the volume label, tick Quick Format and click Finish.
5. Open File Explorer and check the new drive.

Result:

A new partition was created and formatted as NTFS and is visible in File Explorer.

WEEK 9: Monitor and Manage Windows

Activity 9: Troubleshoot slow booting by disabling unnecessary start-up applications

Aim:

To troubleshoot a slow-booting system by disabling unnecessary start-up applications and services.

Procedure:

1. Press Ctrl + Shift + Esc to open Task Manager and go to the Startup tab.
2. Check the Startup impact column and right-click the unwanted high-impact applications and select Disable.
3. Press Win + R, type msconfig, open the Services tab, tick Hide all Microsoft services and disable the unnecessary services.
4. Restart the system and observe the boot time.

Result:

Unnecessary start-up applications and services were disabled and the boot time improved.

WEEK 10: Network Configuration and Management

Activity 10: Configure IP settings and test connectivity using network commands

Aim:

To configure IP and DNS settings and test the network using ping and tracert.

Procedure:

1. Press Win + R, type ncpa.cpl, right-click the adapter and open Properties, then Internet Protocol Version 4.
2. Enter the IP address, subnet mask, default gateway and DNS server (or select Obtain automatically).
3. In Command Prompt run ipconfig /all and note the settings.
4. Run ping 8.8.8.8 and tracert google.com to test connectivity and see the route.
5. Run ipconfig /flushdns to clear the DNS cache.

Result:

The IP settings were configured and the network connectivity was tested using ping and tracert.

WEEK 11: Network Monitoring and Virtual Networks

Activity 11: Install a VM (hypervisor) and document its network utilization

Aim:

To install a virtual machine and document its network utilization.

Procedure:

1. Install VirtualBox or Hyper-V on the host system.
2. Create a new virtual machine and allocate RAM, CPU and disk space.
3. Attach the OS ISO file and install the guest operating system.
4. Set the network adapter to NAT or Bridged and run ipconfig and ping inside the guest OS.
5. Open Task Manager or Resource Monitor on the host and note the network utilization of the VM.

Result:

The VM was installed and its network utilization was documented.

WEEK 12: E-Waste Management

Activity 12: Find the latest e-waste regulations and policies

Aim:

To find the latest e-waste regulations and policies of the Government of India and Karnataka.

Procedure:

1. Open https://greene.gov.in and search for the e-waste rules and policies.
2. Open https://kspcb.karnataka.gov.in and find the e-waste notifications of Karnataka.
3. Note the name of the rules (for example, the E-Waste Management Rules), the year, and the duties of producers, recyclers and consumers.
4. Write a short summary of the points.

Result:

The latest e-waste regulations of the Government of India and Karnataka were found and documented.

WEEK 13: Mini Project Activity - Layout Management

Activity 13: Lab setup plan

Aim:

To develop a lab setup plan including hardware procurement, installation, electrical safety and cybersecurity.

Procedure:

1. Decide the number of systems, the seating arrangement and the lab area, and draw the layout.
2. Prepare the list of hardware to purchase (PCs, switch, router, printer, UPS) with approximate cost.
3. Plan the installation of the systems, network cabling and the operating system and software.
4. Plan electrical safety: proper earthing, MCB, surge protection, UPS and a fire extinguisher.
5. Plan cybersecurity: firewall, antivirus, user accounts with passwords, updates and backup.

Result:

A complete lab setup plan was prepared covering the layout, hardware, installation, electrical safety and cybersecurity.

Student Signature	Staff Signature	HOD Signature